Sequential Speed vs Random IOPS

What Really Makes an SSD Feel Fast?

When shopping for an SSD, you'll often see impressive numbers like **7,000 MB/s Read Speed**. While these figures are important, they don't tell the whole story.

A truly fast SSD combines **high sequential speeds** with **excellent random IOPS**, delivering a smooth and responsive computing experience.

Understanding the Difference

Sequential Read & Write

Best for large files

Sequential performance measures how quickly an SSD reads or writes large blocks of continuous data.

Think of it as **reading a book from the first page to the last**—smooth, continuous, and efficient.

If you're transferring a **100GB movie**, sequential speed is the most important performance metric.

Random Read & Write (IOPS)

The Secret Behind a Responsive PC

Random performance measures how quickly an SSD can access thousands of small files stored in different locations.

Think of it as **opening random pages in a huge book** instead of reading it from beginning to end.

This is much more challenging—and much more important for everyday computing.

Random performance is measured in **IOPS (Input/Output Operations Per Second).**

Example:

- Random Read: **1,000,000 IOPS**
- Random Write: **850,000 IOPS**

Why Random Performance Matters More

Many SSDs advertise impressive sequential speeds like **7,000 MB/s**, but still feel slower in daily use.

Making It Easy

- Imagine driving a car.

Sequential Speed is like the **top speed on a highway**.

Random IOPS is like how well the car handles **busy city traffic with constant stops and turns**.

A sports car with an incredible top speed isn't always the quickest in city traffic.

The same applies to SSDs:

High sequential speeds look impressive, but strong random performance is what makes your computer feel truly fast every day.